

MATH 332 Linear Algebra: Theory and Applications

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Quiz 1

Thursday, September 10, 2026

Instructions:

- i. This quiz has **one question**, worth **20 points**.
- ii. The time allowed is **15 minutes**. Approved testing accommodations apply.
- iii. This quiz is closed book, but you may use up to **4** double-sided letter-sized sheets of notes.
- iv. No calculators or other electronic devices are allowed. Silence and put devices in your bag under your desk or at the front of the room.
- v. Work individually. Do not communicate with other students or share materials. Direct questions to the instructor.
- vi. Show all your work to justify your answer. Answers without adequate justification will not receive credit. Simplify fractions and leave other expressions in exact form.
- vii. Print your name clearly on each answer sheet.

I understand these instructions and have not relied on any help for this exam beyond what is allowed, nor provided any help to anyone else beyond what is allowed.

Score

Signature

Date

Printed Name

1. (20 points) Consider the system

$$\begin{aligned}x + y + z &= 6, \\2x + 3y + z &= 11, \\2y + 3z &= 13.\end{aligned}$$

- a. (14 points) Write the system as an augmented matrix. Use forward elimination and back substitution to solve it. Show your row operations.

Answer: The augmented matrix and forward elimination are

$$\begin{aligned}\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 11 \\ 0 & 2 & 3 & 13 \end{array} \right] &\xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -1 \\ 0 & 2 & 3 & 13 \end{array} \right] \\ &\xrightarrow{R_3 \leftarrow R_3 - 2R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 5 & 15 \end{array} \right].\end{aligned}$$

Back substitution gives

$$\begin{aligned}5z = 15 &\implies z = 3, \\y - z = -1 &\implies y = 2, \\x + y + z = 6 &\implies x = 1.\end{aligned}$$

Thus the unique solution is $\boxed{(x, y, z) = (1, 2, 3)}$. There is a pivot in each variable column, so there are no free variables.

- b. (6 points) Replace exactly one equation in the original system with a new linear equation so that the resulting system has no solution. Keep the other two equations unchanged, and briefly justify your answer.

Answer: One possible choice is to replace the third equation with

$$3x + 4y + 2z = 18.$$

Adding the unchanged first two equations gives $3x + 4y + 2z = 17$. These two conclusions contradict each other, so the new system has no solution. Equivalently, subtracting the first two rows from the new third row gives $[0 \ 0 \ 0 \mid 1]$, or $0 = 1$.

Other valid replacements are acceptable. For example, replacing the third equation with $x + y + z = 7$ directly contradicts the first equation.

Quiz 1 Scores (out of 20)

Number of Students: 19, Minimum: 3, Maximum: 20, Mean: 15.9, Median: 18

Standard Deviation: 5.3, Quartiles (Q1, Q3): (14, 20)

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2|00000000
1|14446889
0|369
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